

DIGITAL EVIDENCE INTEGRITY VERIFICATION

23MAT112 Mathematics for Intelligent Systems 2

24AIM111 Introduction to data structures & algorithms

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PROBLEM STATEMENT

Digital evidence is crucial in legal and forensic investigations, but it is vulnerable to tampering, unauthorized modifications, and corruption. Ensuring the integrity, authenticity, and chain of custody of digital evidence is critical for maintaining its admissibility in court.

OUR VISION

To develop a secure and user-friendly digital evidence verification platform that ensures the integrity of uploaded documents by generating unique hash values and storing them on a blockchain. Any unauthorized modifications will trigger an alert, ensuring trust, transparency, and immutability in digital forensics and legal processes.

Hashing Algorithm — SHA-512

SHA-512 does its work in a few stages. These stages go as follows:

- Input formatting
- Hash buffer initialization
- Message Processing
- Output

WHY INTEGRATING HASHING WITH BLOCK CHAIN IS BETTER?

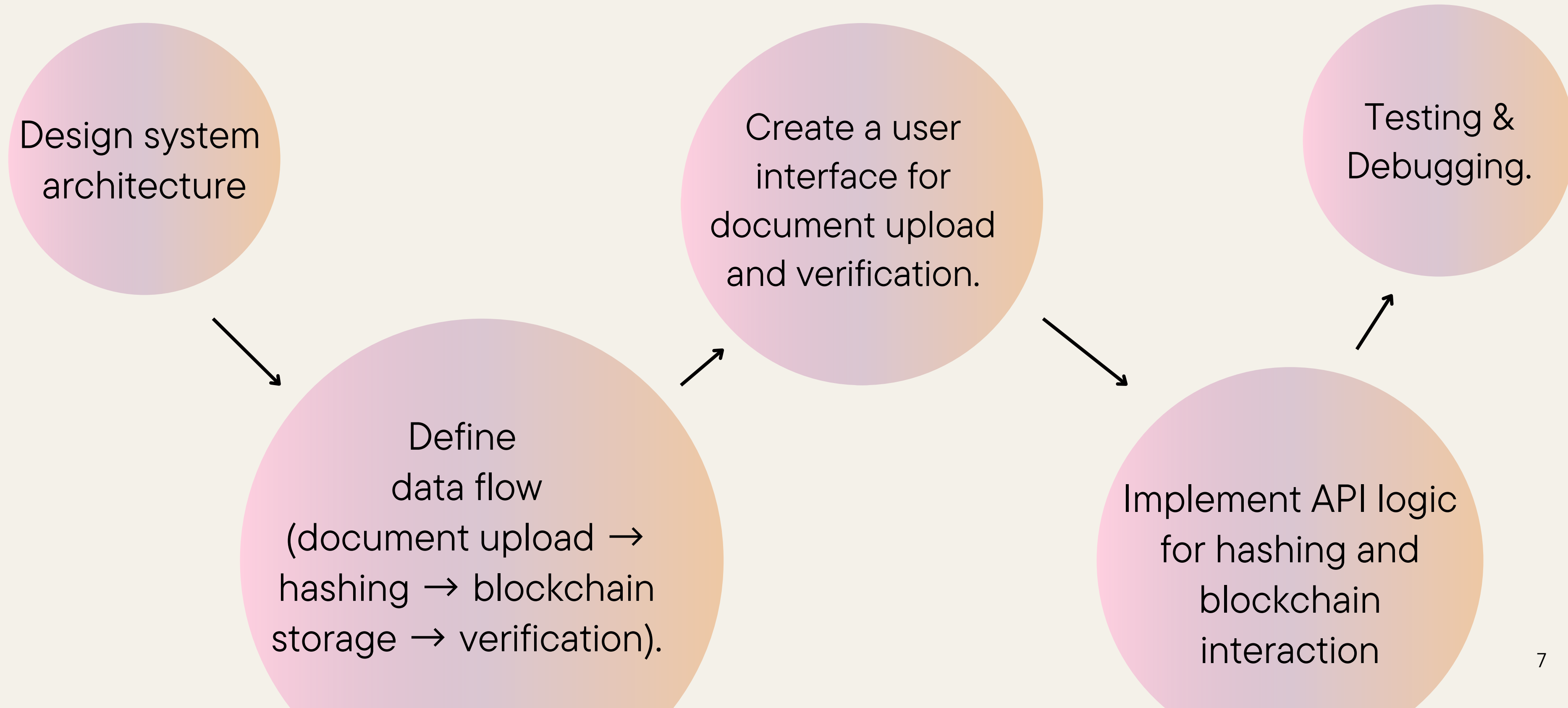
**Immutability &
Tamper Detection**

**Decentralized & Trustless
Verification text**

LITERATURE REVIEW

Title	Methodology	Key takeaways
Evaluation of Security Methods for Ensuring the Integrity of Digital Evidence	Used ITSEC evaluation criteria to evaluate security methods for protecting the integrity of digital evidence	SHA-512 with Digital Signatures should be preferred for integrity protection
Significance of Hash Value Generation in Digital Forensic: A Case Study	Focuses on some case study of hash value generated in digital hard drive for forensic point of view	Role of hash value is demonstrated in this research work with the help of different cases involved in data tampering is analyzed and verified
Potential applicability of blockchain technology in the maintenance of chain of custody in forensic casework	Highlight the potential use of block chain technology to preserve chain of custody	Perfect solution for maintaining and tracing chains of custody as it enforces integrity, transparency, authenticity, security, and auditability by design
Forensic Use of Hash Values and Associated Hash Algorithms	The technical choices made for the uses of hash values are described	Potential risks of using hashing functions for integrity checks

METHODOLOGY



Relevance

Mathematics for Intelligent Systems 2

Linear Algebra & Matrices – SHA-256 and other cryptographic functions rely on matrix-based transformations.

Probability & Random Variables – Understanding randomness in hash functions, entropy, and collision resistance.

Optimization Techniques – Efficient storage and retrieval of hashes

Introduction to data structures and algorithms

Hashing (SHA-512) – used for ensuring data integrity.

Linked Lists (Blockchain Structure) – Each block stores a hash of the previous block, forming a linked list.

Tree Structures (Merkle Trees) – Used in blockchain

Sorting & Searching – Hash tables or binary search.

Thank You